

Propositional Logic: Proof Theory for Infinite Sequents

Extended Completeness Theorem

Logic for Multiagent Systems

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1 Introduction

A central question in formal logic is determining whether a set of premises logically guarantees a conclusion. This is formalized as a sequent $\Gamma \rightarrow \Delta$: if all formulas in Γ are true, at least one formula in Δ must be true. The standard completeness theorem applies when Γ and Δ are finite. This report proves the Extended Completeness Theorem, which extends this result to countably infinite sequences.

The proof relies on the Search Procedure, an algorithm that builds a deduction tree by decomposing formulas using the rules of the Gentzen System G. To handle infinite input without stalling on the initial elements, the algorithm feeds in exactly one new formula per round using two queues. The procedure guarantees one of two outcomes: either the tree closes, producing a formal proof, or it grows infinitely and an infinite path through it produces a Hintikka set that encodes a falsifying valuation. Because neither outcome can be avoided, the system is proven to be complete.

2 Syntax and Semantics

Definition 2.1 (Atomic Propositions). Let $\text{VAR} = \{P_0, P_1, P_2, \dots\}$ be a countably infinite set of atomic propositional symbols.

Definition 2.2 (Formulas). Well formed formulas are built from atoms in VAR using the connectives $\neg$, $\wedge$, $\vee$, and $\rightarrow$.

Definition 2.3 (Valuation). A valuation is a function $v : \text{VAR} \rightarrow \{0, 1\}$, where 1 represents true and 0 represents false. This function extends to all formulas by the standard truth tables.

Definition 2.4 (Infinite Sequent). A sequent is an expression $\Gamma \rightarrow \Delta$, where $\Gamma = A_1, A_2, \dots$ and $\Delta = B_1, B_2, \dots$ are sequences of formulas, possibly countably infinite.

Definition 2.5 (Validity and Falsifiability). A valuation v *falsifies* $\Gamma \rightarrow \Delta$ if v makes every formula in Γ true and every formula in Δ false. The sequent

is *falsifiable* if such a v exists. The sequent is *valid*, written $\models \Gamma \rightarrow \Delta$, if no such v exists.

Definition 2.6 (Axiom). A sequent is an *axiom* if the same atomic formula P appears on both the left side and the right side. An axiom is always valid and requires no further proof.

3 The Gentzen System G

The Gentzen System G is a formal proof system for sequents. Each rule reduces a sequent to simpler subgoals. Rules are read bottom-up during tree construction.

3.1 Conjunction

$$\frac{\Gamma, A, B \rightarrow \Delta}{\Gamma, A \wedge B \rightarrow \Delta} (\wedge L) \qquad \frac{\Gamma \rightarrow \Delta, A \quad \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, A \wedge B} (\wedge R)$$

3.2 Disjunction

$$\frac{\Gamma, A \rightarrow \Delta \quad \Gamma, B \rightarrow \Delta}{\Gamma, A \vee B \rightarrow \Delta} (\vee L) \qquad \frac{\Gamma \rightarrow \Delta, A, B}{\Gamma \rightarrow \Delta, A \vee B} (\vee R)$$

3.3 Implication

$$\frac{\Gamma \rightarrow \Delta, A \quad \Gamma, B \rightarrow \Delta}{\Gamma, A \rightarrow B \rightarrow \Delta} (\rightarrow L) \qquad \frac{\Gamma, A \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, A \rightarrow B} (\rightarrow R)$$

3.4 Negation

$$\frac{\Gamma \rightarrow \Delta, A}{\Gamma, \neg A \rightarrow \Delta} (\neg L) \qquad \frac{\Gamma, A \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A} (\neg R)$$

4 The Search Procedure

Definition 4.1 (Search Procedure). Given $\Gamma \rightarrow \Delta$, initialize $L = \text{tail}(\Gamma)$, $R = \text{tail}(\Delta)$. The root node contains $\text{head}(\Gamma) \rightarrow \text{head}(\Delta)$. Here, head retrieves the first element of a sequence and tail retrieves everything after the first element.

At each step, apply *expand* to every unfinished leaf $\Gamma' \rightarrow \Delta'$:

1. Apply all applicable Gentzen rules to fully decompose $\Gamma' \rightarrow \Delta'$ into child nodes.
2. Append $\text{head}(L)$ to the left side and $\text{head}(R)$ to the right side of each new leaf.
3. Set $L \leftarrow \text{tail}(L)$, $R \leftarrow \text{tail}(R)$.

A leaf is *closed* if the same atom appears on both sides (if it is an axiom). The procedure terminates when all leaves are closed, producing a *closed tree* (a formal proof).

Since $\text{head}(L)$ and $\text{head}(R)$ advance at every step, every formula in Γ and Δ is eventually introduced. If $\Gamma \rightarrow \Delta$ is infinite, any non-axiom leaf is also unfinished (since L or R cannot be empty), so the procedure either terminates with a closed tree or builds an infinite tree.

5 Examples

5.1 Example 1: Valid Sequent

Let the premises be $P_0, (P_0 \rightarrow P_1), (P_1 \rightarrow P_2), \dots$ and the conclusions be Q, P_3 .

The root sequent is $P_0 \rightarrow Q$. The queue $L = (P_0 \rightarrow P_1), (P_1 \rightarrow P_2), \dots$ and $R = P_3$.

At each round, rule $(\rightarrow L)$ is applied to the implication at the head of L . This splits the current leaf into two branches. The left branch always contains the same atom on both sides of the sequent, making it an axiom

that closes immediately. The right branch receives the next element from L and continues to the next round. After four rounds, all branches are closed. The complete proof tree is shown in Figure 1.

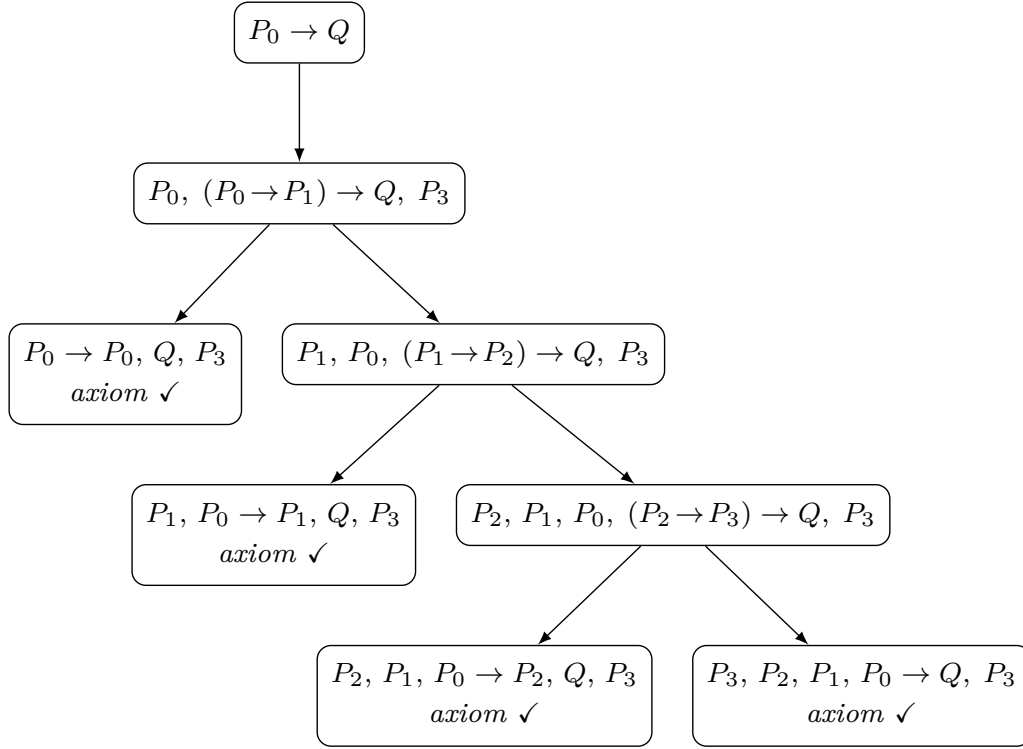


Figure 1: Proof tree for Example 1. Every branch ends in a closed axiom.

5.2 Example 2: Invalid Sequent

Remove P_3 from the conclusions. Now the right queue R is empty and the conclusion is just Q .

The algorithm runs as before, but the rightmost branch never closes. After round n , the open leaf is $P_n, \dots, P_0 \rightarrow Q$. This branch extends infinitely, never reaching an axiom.

Collecting all formulas along this infinite branch: $U = \{P_0, P_1, \dots, (P_0 \rightarrow P_1), (P_1 \rightarrow P_2), \dots\}$ and $V = \{Q\}$. Setting all P_i to true and Q to false falsifies the sequent. This counterexample is made precise by the notion of a

Hintikka set.

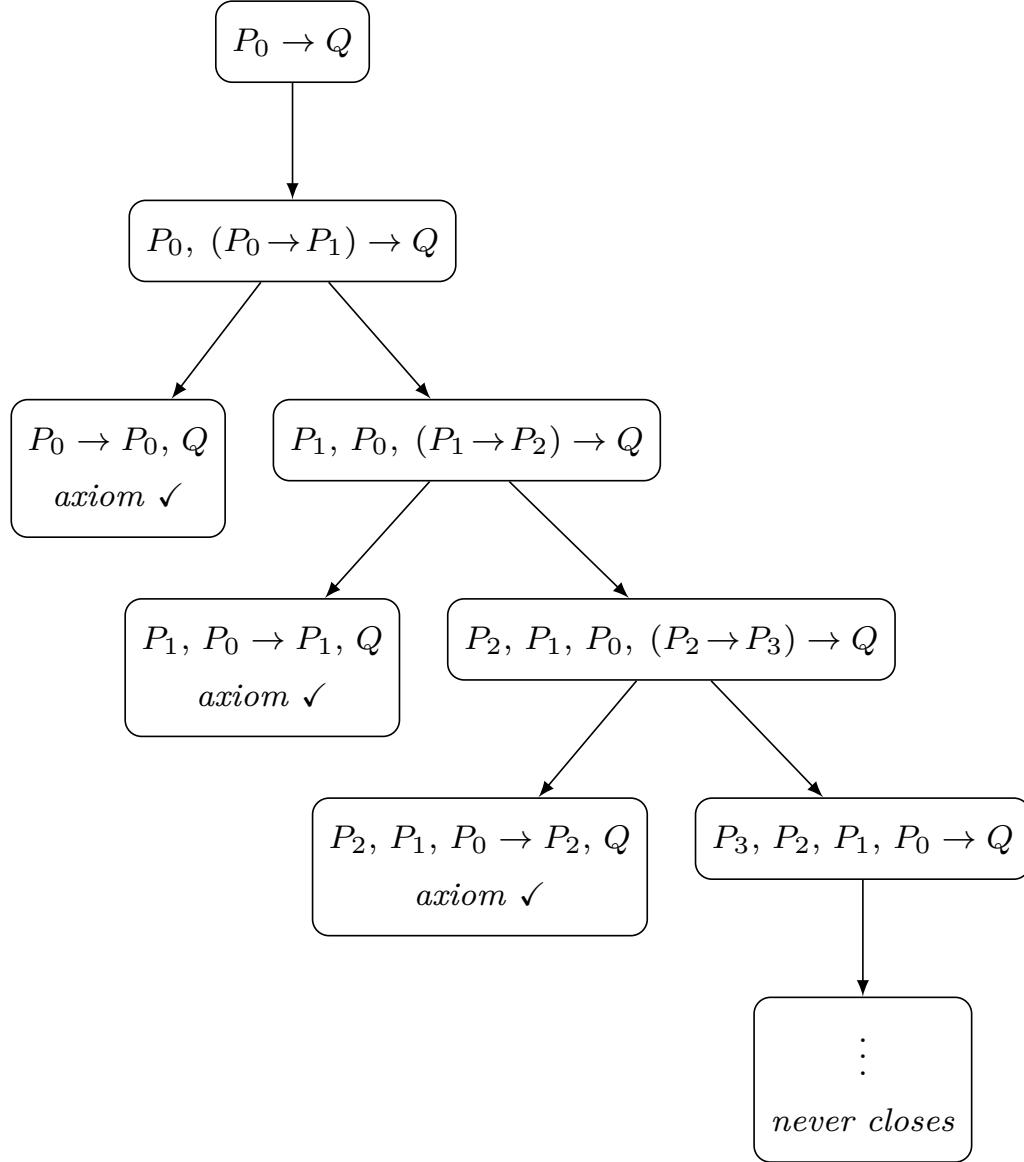


Figure 2: Partial proof tree for Example 2. The rightmost branch extends infinitely and never reaches an axiom.

6 Koenig's Lemma

Lemma 6.1 (Koenig's Lemma). *Every finite-branching tree (where each node has finitely many children) with infinitely many nodes contains an infinite path.*

Proof. Define the path inductively. Start at the root. Since the root has finitely many children but the tree is infinite, at least one child subtree must be infinite. Select the root of the leftmost such subtree. Repeat: at each node, finitely many children exist and the current subtree is infinite, so at least one child subtree is again infinite. Selecting the leftmost such child at each step defines an infinite path. $\square$

Since the Gentzen rules produce at most two children per node, the search tree is always finite-branching. Therefore, if the procedure builds an infinite tree, Koenig's Lemma guarantees an infinite path.

7 Signed Formulas

Definition 7.1 (Signed Formula). A signed formula is an expression TA or FA where A is a proposition. Given $\Gamma \rightarrow \Delta$, its signed set is $\{TA \mid A \in \Gamma\} \cup \{FB \mid B \in \Delta\}$. The *conjugate* of TA is FA and vice versa. A valuation satisfies a signed set if it makes every signed formula in it true.

Signed formulas are classified by how they decompose.

Type a (conjunctive: true if and only if both components are true):

Formula	Component 1	Component 2
$T(X \wedge Y)$	TX	TY
$F(X \vee Y)$	FX	FY
$F(X \rightarrow Y)$	TX	FY
$T(\neg X)$	FX	FX
$F(\neg X)$	TX	TX

Type b (disjunctive: true if and only if at least one component is true):

Formula	Component 1	Component 2
$F(X \wedge Y)$	FX	FY
$T(X \vee Y)$	TX	TY
$T(X \rightarrow Y)$	FX	TY

8 Hintikka Sets

Definition 8.1 (Hintikka Set). A set S of signed formulas is a *Hintikka set* if:

- (H1) No atomic signed formula and its conjugate are both in S .
- (H2) If a type a formula is in S , both its components are in S .
- (H3) If a type b formula is in S , at least one of its components is in S .

Lemma 8.2. *If the search procedure does not produce a closed tree, a Hintikka set can be extracted from the tree.*

Proof. If the tree is finite with a non-axiom leaf, select a path to that leaf. If the tree is infinite, select an infinite path guaranteed by Koenig's Lemma. Let U be the union of left sides and V the union of right sides along the chosen path. Define $S = \{TA \mid A \in U\} \cup \{FB \mid B \in V\}$.

H1: If both TP and FP were in S , some sequent on the path would have P on both sides, making it an axiom, which contradicts the choice of path.

H2: If a type a formula $A \in S$ is processed by the expansion step, the corresponding Gentzen rule produces a single successor node that contains both components A_1 and A_2 . Since the path passes through this successor, both A_1 and A_2 are in S .

H3: If a type b formula $B \in S$ is processed, the corresponding Gentzen rule produces two successor nodes: B_1 goes to the left child and B_2 goes to the right child. The path passes through exactly one of these, so at least one of B_1, B_2 is in S .

Therefore S is a Hintikka set. □

Lemma 8.3. *Every Hintikka set S is satisfiable.*

Proof. Define $v(P) = 1$ if $TP \in S$, $v(P) = 0$ if $FP \in S$, and $v(P) = 1$ otherwise. Condition H1 ensures v is well defined. By induction on the number of connectives: if a type a formula is in S , H2 ensures both components are in S , and by induction v makes both true, therefore v makes the formula true. If a type b formula is in S , H3 ensures at least one component is in S , and by induction v makes that component true, therefore v makes the formula true. $\square$

9 The Extended Completeness Theorem

Theorem 9.1 (Gallier, Theorem 3.5.1). *Given a sequent $\Gamma \rightarrow \Delta$, exactly one of the following holds:*

1. $\Gamma \rightarrow \Delta$ is valid: the search procedure terminates with a closed tree, and finite subsequences $C_1, \dots, C_m$ of Γ and $D_1, \dots, D_n$ of Δ exist such that $C_1, \dots, C_m \rightarrow D_1, \dots, D_n$ is provable.
2. $\Gamma \rightarrow \Delta$ is falsifiable: the procedure either produces a finite tree with a non-axiom leaf (if $\Gamma \rightarrow \Delta$ is finite) or runs forever (if $\Gamma \rightarrow \Delta$ is infinite), and a falsifying valuation is obtained from a path in the tree.

Proof. **Case 1: The procedure terminates with a closed tree T .** Let $C_1, \dots, C_m$ and $D_1, \dots, D_n$ be the formulas introduced from L and R during the run. A proof tree T' for $C_1, \dots, C_m \rightarrow D_1, \dots, D_n$ is obtained from T by inserting the introduced formulas into every node above their introduction depth and removing duplicates. The tree T' uses the same rules as T and constitutes a valid proof. A valid subsequence implies validity of the full sequent.

Case 2: The procedure does not terminate with a closed tree. If $\Gamma \rightarrow \Delta$ is finite, the tree is finite with a non-axiom leaf; select a path to it. If $\Gamma \rightarrow \Delta$ is infinite, every non-axiom leaf is unfinished (the queues L and R cannot be empty while Γ or Δ is infinite), so the tree is infinite and Koenig's Lemma provides an infinite path. In both cases, by Lemma 8.2 the path produces a Hintikka set S , and by Lemma 8.3 a satisfying valuation v

exists. Since v makes all formulas in $U \supseteq \Gamma$ true and all formulas in $V \supseteq \Delta$ false, it falsifies $\Gamma \rightarrow \Delta$. $\square$

Theorem 9.2 (Extended Completeness Theorem, Gallier Theorem 3.5.2). *For every possibly infinite sequent $\Gamma \rightarrow \Delta$, the sequent is valid if and only if it is provable.*

Proof. Soundness (provable $\Rightarrow$ valid) holds because every Gentzen rule preserves validity. Completeness (valid $\Rightarrow$ provable) follows from Theorem 9.1: if $\Gamma \rightarrow \Delta$ is valid, Case 2 cannot apply, so the procedure must terminate with a closed tree, producing a proof. $\square$

10 Consequences of Completeness

Definition 10.1 (Satisfiability of a Set). A set Γ of propositions is *satisfiable* if there exists a valuation v such that $v \models A$ for every $A \in \Gamma$.

Corollary 10.2 (Compactness Theorem). *For any possibly infinite set of propositions Γ , if every finite subset of Γ is satisfiable, then Γ is satisfiable.*

Proof. Suppose Γ is not satisfiable. Then the sequent $\Gamma \rightarrow$ (with empty right side) is valid. By Theorem 9.1, the search procedure terminates with a closed tree, and there exists a finite subsequence $C_1, \dots, C_m$ of Γ such that $C_1, \dots, C_m \rightarrow$ is provable, therefore valid. This means $\{C_1, \dots, C_m\}$ is not satisfiable, contradicting the hypothesis. Therefore Γ is satisfiable. $\square$

Definition 10.3 (Consistency). A set Γ of propositions is *consistent* if there exists some proposition B such that the sequent $\Gamma \rightarrow B$ is not provable. Otherwise, Γ is *inconsistent*.

Theorem 10.4 (Model Existence Theorem). *If a set Γ of propositions is consistent, then it is satisfiable.*

Proof. Assume Γ is unsatisfiable. Then for every proposition B , the sequent $\Gamma \rightarrow B$ is valid, since there is no valuation making all of Γ true. By Theorem 9.1, for every such B there is a finite subsequence $A_1, \dots, A_p$ of Γ

such that $A_1, \dots, A_p \rightarrow B$ is provable. Since this holds for every B , the set Γ is inconsistent, contradicting the hypothesis. Therefore Γ must be satisfiable. $\square$

Lemma 10.5 (Consistency Lemma). *If a set Γ of propositions is satisfiable, then it is consistent.*

Proof. Let v be a valuation such that $v \models A$ for every $A \in \Gamma$. Suppose for contradiction that Γ is inconsistent. Then there is a finite subsequence $A_1, \dots, A_m$ of Γ such that $A_1, \dots, A_m \rightarrow P \wedge \neg P$ is provable for some propositional symbol P . By the soundness of System G, this sequent is valid. Since v makes all of $A_1, \dots, A_m$ true, the valuation v must also make $P \wedge \neg P$ true, which is impossible. Therefore Γ is consistent. $\square$

11 Conclusion

This report has presented the Extended Completeness Theorem for the Gentzen System G, following Gallier [1]. The proof rests on three auxiliary results: Koenig's Lemma, which guarantees an infinite path in the search tree when the procedure does not terminate; the extraction lemma (Lemma 8.2), which produces a Hintikka set from that path; and the satisfiability lemma (Lemma 8.3), which produces a falsifying valuation from any Hintikka set. Together, these results show that if the search procedure fails to close the tree, a concrete falsifying valuation is always obtainable.

References

- [1] Jean H. Gallier. *Logic For Computer Science: Foundations of Automatic Theorem Proving*. 2nd edition. Dover Publications, 2003. Section 3.5.